Air Insight

Data-Driven AQI, Health, and Policy Exploration Tool

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Summary

This project creates an interactive air quality visualization platform allowing users to explore current, historical, and predicted Air Quality Index (AQI) by United States (U.S.) county.

Air pollution is a major, preventable cause of illness and premature death. Existing research has focused on identifying the problem rather than providing actionable solutions. An interactive tool that enables consumers to visualize air quality alongside related factors does not exist today, nor the ability to model how interventions could improve outcomes.

Users may overlay AQI data with other factors, such as electric vehicle (EV) charging stations and air pollution-related health incidents. As a novel feature, users can simulate how policy interventions could impact air quality and predict the future air quality level.

Data

Data for this project was downloaded from the following sources:

Data	Source	Size	Records	Type
Daily AQI readings by county	U.S. Environmental Protection Agency (EPA)	294 MB	125,280	Timeseries (daily)
EV charging stations	U.S. Department of Energy (DOE)	188 MB	537,086	Geolocation
Health indicators	Centers for Disease Control (CDC)	280 MB	99,904	CSV
Population	National Cancer Institute and U.S. Census Bureau	89 MB	358,417	CSV
U.S. ecoregions	U.S. EPA	42 MB	3,114	Geolocation
Custom U.S. county map file	U.S. Census Bureau	842 KB	N/A	GeoJSON
County-state unique identifier list	U.S. Census Bureau	18 MB	3,144	CSV
Daily temperature by county	U.S. EPA	1.11 GB	3,616,978	Timeseries (daily)

Table 1: List of Data Sources

Our team used Google Cloud Platform (GCP) buckets to store and GCP BigQuery to process the data. After uploading the data, we cleaned, normalized, transformed, and aggregated data to query using Microsoft Power BI.

Holt-Winters Exponential Smoothing

Approach

After fitting an additive Holt-Winters model on cleaned county-level AQI data, we analyzed a 12-month forecast with a 95-percent confidence interval for each U.S. County between April 2025 and March 2026. In Figure 1, Baldwin County's air quality showcases clear annual seasonality and moderate month-to-month variability. The decomposition indicates that AQI fluctuations are driven by seasonal effects rather than structural shifts. The baseline level declines steadily until around 2021, after which it begins to rise again in 2022, suggesting a rebound in underlying AQI levels. Similarly, the trend component shows a gradual upward trajectory beginning in 2022, indicating a significant shift toward slightly worsening air quality.

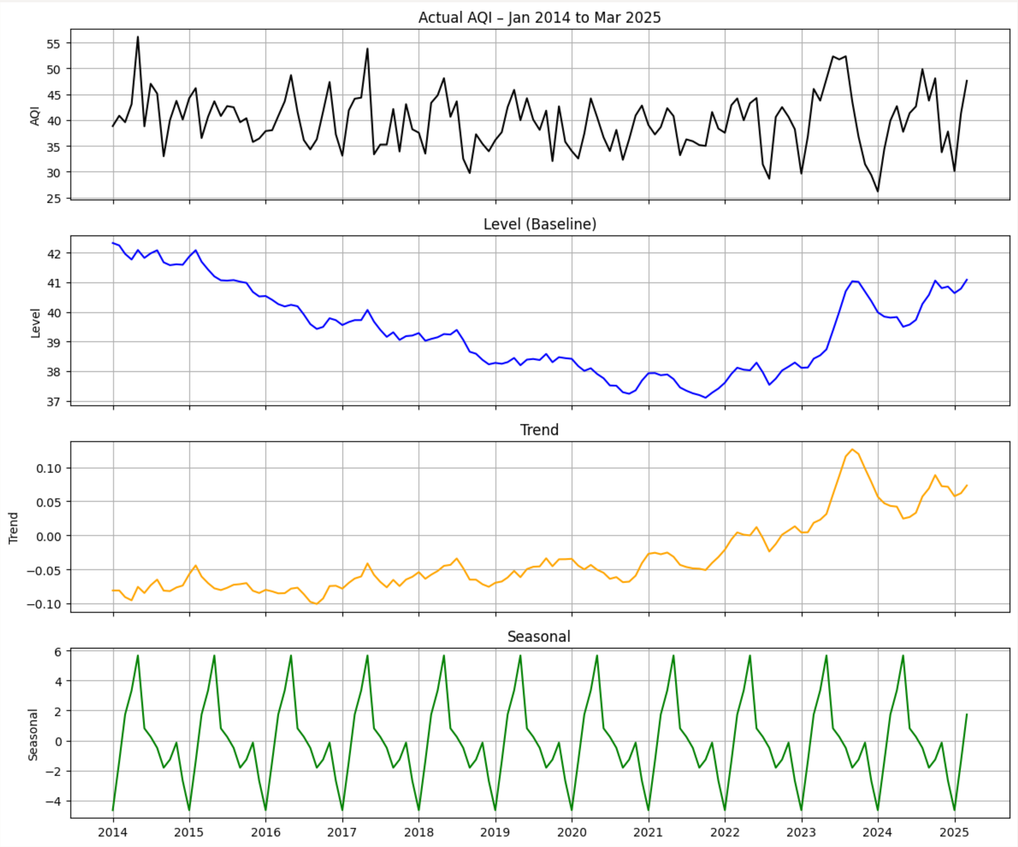

Figure 1: Holt-Winters Temporal Components for Baldwin County, AL

Results

Across the hundreds of counties analyzed, the average forecast performance resulted in a root mean squared error (RMSE) of 6.9 AQI units and a mean absolute error (MAE) of 5.5 AQI units. The relationship between RMSE and MAE is useful in understanding the severity of outliers, and their relative proximity suggests that the errors are reasonably consistent. Given that AQI categories spans 50 points, the model's forecasts typically remain within the same air-quality category as the actual observations. The results suggest that the Holt-Winters model provides stable, interpretable forecasts with a level of precision suitable for practical air-quality visualization and planning purposes.

Electric Vehicle (EV) Intervention Modeling

Approach

The EV analysis merges EV station data with AQI, demographic, political, and geographic datasets. AQI was normalized by log-population density and plotting the normalized AQI against EV station density results in a moderate negative correlation, indicating that more EV charging stations may lower AQI. Our team wanted to ensure that we are considering that correlation does not always indicate causation, and researched additional data points that may affect air quality. For example, EV station density is also found to be correlated to Democratic-voter density, where counties with a higher percentage of Democratic voters have more EV stations. This illustrates the belief that counties with stronger environmental policy preferences both adopt EV infrastructure more rapidly and implement other pollution-reducing policies.

To better isolate the relationship between EV charging infrastructure and air quality, we created a two-way fixed-effects model on the 2014–2024 dataset. County fixed effects account for all characteristics that do not change over time, while year fixed effects remove nationwide influences like wildfire seasons or unusually warm years.

Across the different versions of the model, the coefficient on EV station density remains small but consistently negative.

Results

Overall, EV charging infrastructure highlights a smaller but meaningful relationship with improved air quality, and the analysis reinforces the fixation on meteorological, geographical, and political factors that are prevalent in past literature. Figure 2 displays projected AQI paths for Mono County, California, under the growth scenarios.

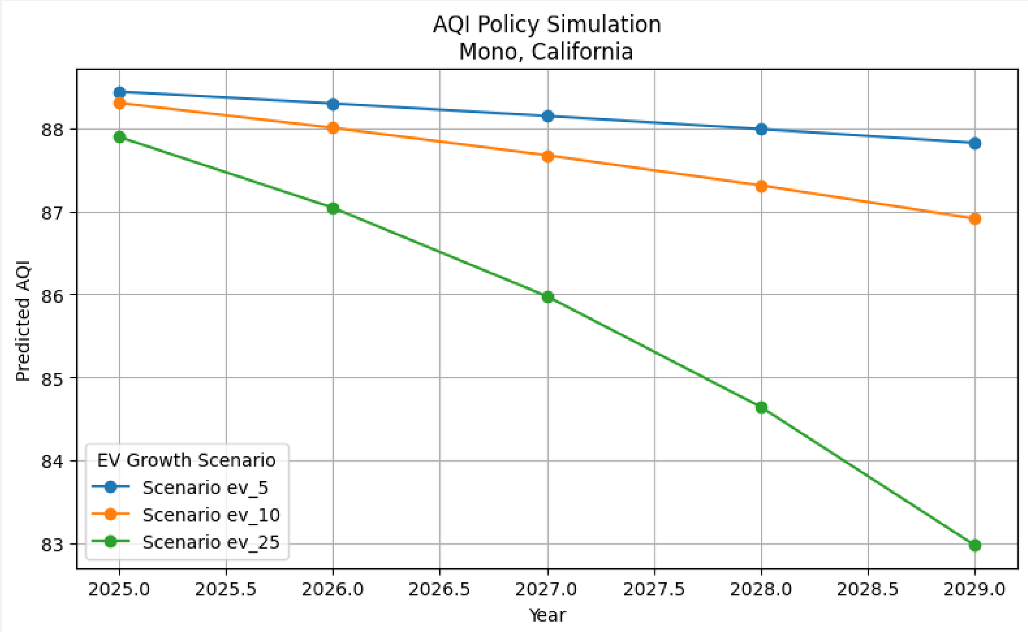

Figure 2: EV Intervention Modeling Results

Data Visualization

Approach

The data visualizations are developed in Microsoft Power BI, integrated directly with Google BigQuery data views. The dashboard features three tabs, each with interactive maps for (1) Pollution-related health indicators, (2) AQI forecasts, and (3) simulated EV infrastructure effects on AQI. When a user hovers over a county, the tooltip displays both the actual and predicted AQI for that location. All views allow the user to interact with the data and/or the algorithms through clicking and hovering actions.

Results

Our final product delivers a user interface that allows a user to interact with eight cleaned (Table 1) and two algorithm-based (AQI forecasts and AQI predictions based on EV station increases) datasets. Users can use the application to view current and forecasted future AQI based on U.S. county, and they can also simulate how policy interventions such as additional EV stations may improve air quality for a given state or county, as shown in Figures 3, 4, and 5.

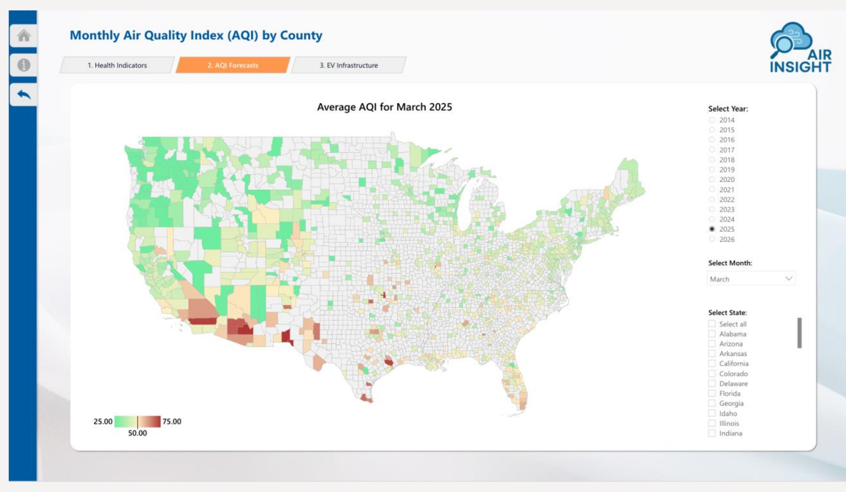

Figure 3: Historical and Forecasted AQI by U.S. County

Figure 4: Dynamic Tooltip with Forecasted AQI

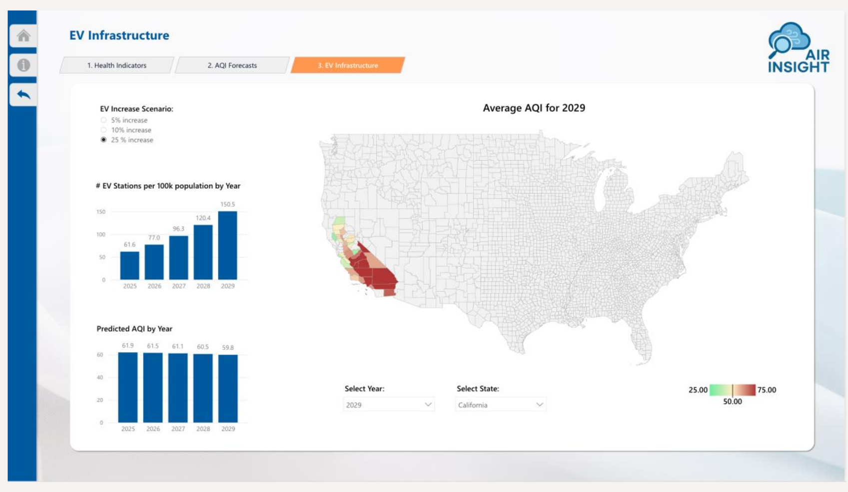

Figure 5: Simulated AQI Based on EV Station Increases

Conclusion

Discussion

The project objective was to create an interactive air quality visualization to enable policymakers and the public to make informed, data-driven decisions and explore interventions that could improve future air quality outcomes.

Even facing challenges and potential for expansion, our current dashboard and predictive algorithms provide significant impact to the general public. The ability to not only visualize past AQI but also view predicted future AQI allows people to take air quality into account when determining where to live. It also allows both the general public and policymakers the ability to simulate how air quality may change if policy decisions, such as increased EV infrastructure, were to go into effect.

Further Research & Analysis

If research continued on this project, we would look to address the following items:

1. *Additional EV Data*: EV charging stations remain quite sparse, therefore we would like to expand our EV data to include EV ownership by county in addition to existing infrastructure.
2. *Political Views*: While we began to explore the relationship between EV stations and political ideology, we would like to dig deeper, specifically looking at the strength of air-quality regulation policies that underly the relationship of AQI with political ideology.
3. *Expanded Meteorological Variables*: Temperature is important for explaining AQI patterns, but is not the only important meteorological variable (wind, humidity, etc.). By adding more weather factors, we gain deeper insights into how much EV stations matter relative to further weather changes.